

TABLE II
Technological Application

Technologies	Description	Short term (2010)	Mid term (2015)	Long term (2020+)
System architecture	An architecture defines the structure of system components, their interrelationships and the principles governing their design and evolution over time. Robot architectures should adapt approaches from neighbouring industries (telecom, aerospace and automotive), focussing on physical human interaction.	Hierarchical architectures running on a single system; architecture may use multiple cores for specific purposes	Hybrid or layered, service-oriented architectures; loosely coupled distributed modules (real-time agents)	Component compositionality and self-configuration; globally distributed, resource-aware architectures
System engineering tools	These are tools for designing a robot system (hardware and software) including simulation of its dynamic properties and deployment. Robotics can benefit from the aerospace, automotive, manufacturing systems, games and defence industries.	Separate tools exist to aid the design of aspects of robot and application; simplistic models, which cannot be linked	Integrated tool chain for design of robot and application (easily extendable); dynamic robot models	Integrated tool chain to custom-build robots; detailed, easy-to-use dynamic models for robot and environment
Cooperating robots and ambient intelligence	In this field, the desired collective behaviour emerges from robot-robot interactions and their interactions with the environment. With the exception of communication and sensor networks, this area is driven by robotics.	Teams of robots; centralised control and communication; tasks specified for each individual robot; use of common map	Distributed control; inter-agent communication; task specified for team; games and swarm theories are applied	Cooperation without explicit representation of action; skill-based or learning-based automation
(Real-time) Communication	This field is concerned with hardware and software communication within the system's time constraints in the context of its architecture. The transfer of solutions from aerospace and the consumer electronics industry to robotics is non-trivial and has to be supported. Open frameworks for software and hardware also play an important role.	Numerous specialised protocols; Ethernet-based communication starts to take over as de facto standard	New protocols using ontologies, logic, probabilistic or geometric models, rule sets, etc. ...	Components can figure out each others' protocols; components negotiate required quality of service
Human-machine interface	Interfaces enable humans and robots to communicate with each other using a variety of channels. Human-machine and human-computer interfaces need to be extended to robotics and physical interaction.	Mostly graphical or text-based interfaces; few haptic devices and use of human interaction channels; touch interfaces	Human interaction channels, which humans have to learn; some telepresence; haptic input devices; learning interfaces	Interaction using human channels utilising cognitive approaches; neural interfaces; non-invasive brain interfaces
Safety	Safety considers how to avoid or handle hazardous situations to reduce the severity and likelihood of harm to acceptable levels. Safety methodologies from other domains must be adapted for robotic systems.	Sensor-based physical safety; hardware safety through redundancy; software safety through formal approaches to programming	Model-based hardware and software failure detection and isolation; application safety (explosives, food, medicine, etc)	Predictive failure detection; safe automatic obstacle avoidance; detection of the intention of a person
Actuation	Actuation technologies generate forces and torques to thereby manage the motion of robots. Only specialised parts such as lightweight, compact drives and gears designed for frequent speed and direction changes are driven by robotics.	Mostly electric, pneumatic, or hydraulic motors; lightweight high-density actuators; standard gears	Continuously variable transmissions; ball-socket joints; improved energy-saving and power-weight ratio	High-energy efficiency; safe, powerful actuators; micro-actuation; use of smart materials; powerful pneumatics and hydraulics
End effectors	End effectors enable a robot to interact with and change its environment, eg, by grasping, manipulating and processing objects. Grippers, hands, process tools and tool changers are developed by the robotics community, but the prostheses industry is also a stakeholder.	Task-specific end effectors, especially grippers; mostly pre-programmed or taught grasping strategies; flexibility with tool changers	Multi-finger grippers for a variety of objects; grasps computed online; gripping of human tools	Dexterous hands; grasping of all objects; use of multiple hands; future goal: human dexterity and assembly skills
Locomotion	Locomotion allows a robot to move to a specified location on the ground, in the air, in space, on or under water or inside a living body. Except for biologically inspired locomotion, most aspects of locomotion are driven by other sectors.	Engineering solutions to locomotion; locomotion inside the human body through external force fields	Biomimetic locomotion in/on water and on land; bipedal locomotion in structured environments	Bipedal locomotion in unstructured environments (mostly indoors); energy efficiency; autonomous in-body locomotion